WORKED SOLUTION

STEP II 1998, Question 2

Pure • Binomial Series Approximation

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q2

The key idea

A binomial series converges fast when its argument is small, so we want to write $\sqrt{2}$ (or $\sqrt[3]{2}$) as a simple rational number times $(1+x)^a$ with x tiny. For square roots, $49/50$ is a near miss for 1 that contains a factor of 2; for cube roots we look for a cube that is almost $2\times$ another cube, and $128 = 2\times 4^3$ sits right next to $125 = 5^3$.

Solution

The square root of 2

First we connect the given expression to $\sqrt{2}$. Since $1-\frac{1}{50}=\frac{49}{50}$,

$$\left(1-\frac{1}{50}\right)^{1/2}=\frac{7}{\sqrt{50}}=\frac{7}{5\sqrt{2}}=\frac{7\sqrt{2}}{10},\qquad\text{so}\qquad\sqrt{2}=\frac{10}{7}\left(1-\frac{1}{50}\right)^{1/2}.$$

The binomial series for $(1+x)^{1/2}$ begins

$$(1+x)^{1/2}=1+\tfrac{1}{2}x-\tfrac{1}{8}x^2+\tfrac{1}{16}x^3-\cdots,$$

valid for $|x|<1$. Put $x=-\frac{2}{100}=-2y$ with $y=\frac{1}{100}$. The powers of 2 cancel nicely:

$$\begin{aligned}\left(1-\frac{2}{100}\right)^{1/2}&\approx 1-\tfrac{1}{2}(2y)-\tfrac{1}{8}(2y)^2-\tfrac{1}{16}(2y)^3\\&=1-y-\tfrac{1}{2}y^2-\tfrac{1}{2}y^3\\&=1-0.01-0.00005-0.0000005\\&=0.9899495.\end{aligned}$$

This is why writing $1/50$ as $2/100$ helps: every term becomes a power of 10 times 1 or $\frac{1}{2}$, so the arithmetic is exact decimals.

Hence

$$\sqrt{2}\approx\frac{10\times 0.9899495}{7}=\frac{9.899495}{7}=1.41421357\ldots,$$

and to six decimal places $\sqrt{2} \approx 1.414214$, as required.

The cube root of 2

Ask yourself Which N and a? What makes $(1+N/125)^a$ related to $\sqrt[3]{2}$?

Since $125 = 5^3$, a cube root of $(125+N)/125$ will be $\sqrt[3]{125+N}/5$. We therefore want $125+N$ to be 2 times a perfect cube, with N small. The cubes 2×1^3, 2×2^3, 2×3^3, $2\times 4^3 = 2, 16, 54, 128$ show that 128 is only 3 away from 125. So take $N = 3$ and $a = \frac{1}{3}$:

$$\left(1+\frac{3}{125}\right)^{1/3} = \frac{\sqrt[3]{128}}{5} = \frac{4\sqrt[3]{2}}{5}, \qquad \text{so} \qquad \sqrt[3]{2} = \frac{5}{4}\left(1+\frac{3}{125}\right)^{1/3}.$$

The binomial series for $(1+x)^{1/3}$ begins

$$(1+x)^{1/3} = 1 + \tfrac{1}{3}x + \frac{\frac{1}{3}\cdot(-\frac{2}{3})}{2!}x^2 + \frac{\frac{1}{3}\cdot(-\frac{2}{3})\cdot(-\frac{5}{3})}{3!}x^3 + \cdots = 1 + \tfrac{1}{3}x - \tfrac{1}{9}x^2 + \tfrac{5}{81}x^3 - \cdots.$$

Just as before, a substitution makes the arithmetic clean. Put $x = 3y$ with $y = \frac{1}{125} = 0.008$; then $\frac{1}{3}x = y$, $\frac{1}{9}x^2 = y^2$ and $\frac{5}{81}x^3 = \frac{5}{3}y^3$. So

$$\begin{aligned}\left(1+\frac{3}{125}\right)^{1/3} &\approx 1 + y - y^2 + \tfrac{5}{3}y^3 \\ &= 1 + 0.008 - 0.000064 + \tfrac{5}{3}(0.000000512) \\ &= 1.007936 + 0.00000085333\ldots \\ &= 1.00793685333\ldots.\end{aligned}$$

Multiplying by $\frac{5}{4}$,

$$\sqrt[3]{2} \approx 1.25 \times 1.00793685333\ldots = 1.25992106\ldots,$$

so to six decimal places $\sqrt[3]{2} \approx 1.259921$.

Checks

A calculator gives $\sqrt{2} = 1.41421356\ldots$ and $\sqrt[3]{2} = 1.25992105\ldots$, so both approximations are correct to six decimal places; our values are 1.41421357 and 1.25992107. We checked both series with exact fractions by computer. The first omitted terms give a sense of the error: $-\frac{5}{128}x^4$ contributes about -9×10^{-9} to the $\sqrt{2}$ estimate, and $-\frac{10}{243}x^4$ contributes about -1.7×10^{-8} to the $\sqrt[3]{2}$ estimate. Both are far below the 5×10^{-7} that would change the sixth decimal place.

Common slips

- Forgetting to undo the scaling: the series gives $\frac{7\sqrt{2}}{10}$ and $\frac{4\sqrt[3]{2}}{5}$, not $\sqrt{2}$ and $\sqrt[3]{2}$ themselves.

- Sign errors in the series. With x negative in the first part, all four terms after the 1 are negative. In the second part the signs alternate.
- Rounding too early. Keep at least eight decimal places until the final division or multiplication.

Beyond STEP

Behind this question is Taylor's theorem with a remainder term, which turns "the next term is tiny" into a rigorous bound on the error. It explains why a series with ratio about 0.02 gains roughly one and a half to two decimal places per term. The same trick, finding a rational number p/q with p^n/q^n close to a simple multiple of the target, links to continued fractions and rational approximation in number theory. Davenport's *The Higher Arithmetic* is a good place to read about these, and Abbott's *Understanding Analysis* covers Taylor's remainder carefully.

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